



COLLEGE
OF
MEDICINE

Gestational trophoblastic disease

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Gestational trophoblastic disease. GTD

- **Theoretical skills**
 - Understand the pathogenesis.
 - Have good knowledge of gynaecological diagnosis and management.
 - Have an appreciation of the needs for monitoring and/or medical intervention.
- **Practical skills**
 - Be able to counsel a woman with gestational trophoblastic disease.
 - Be able to recognize the clinical features of gestational trophoblastic disease.
 - Be able to perform evacuation of molar pregnancy under supervision



Gestational trophoblastic disease

- Gestational trophoblastic disease (**GTD**) is an uncommon complication of pregnancy.
- The term 'GTD' describes a group of inter-related diseases, including complete and partial molar pregnancy, choriocarcinoma and placental site trophoblastic tumour, which vary in their propensity for local invasion and metastasis.
- Although persistent GTD (now often termed gestational trophoblastic neoplasia (GTN)) most commonly follows a molar pregnancy, it may also be seen after any type of gestation, including term pregnancy, abortion and ectopic pregnancy.



Risk factors for GTD

- Age : ↑ <16 years, ↑↑↑ >45 years
- Geographic ↑ Asia
- Previous GTD ↑ one episode, ↑↑↑ two or more episodes
- Dietary ↑ carotene deficiency
- Genetic Rare family clusters and repetitive moles
- Maternal age appears to be the most consistent risk factor associated with molar gestation Women who have had a previous mole have an increased
- risk of further molar pregnancies. Following one mole, the risk is less than 2 per cent, but following two molar pregnancies it increases substantially up to one in six; following three moles the risk may be as high as one in two



PATHOLOGICAL FEATURES

Hydatidiform mole may be complete or partial and the histopathological features differ

Complete mole

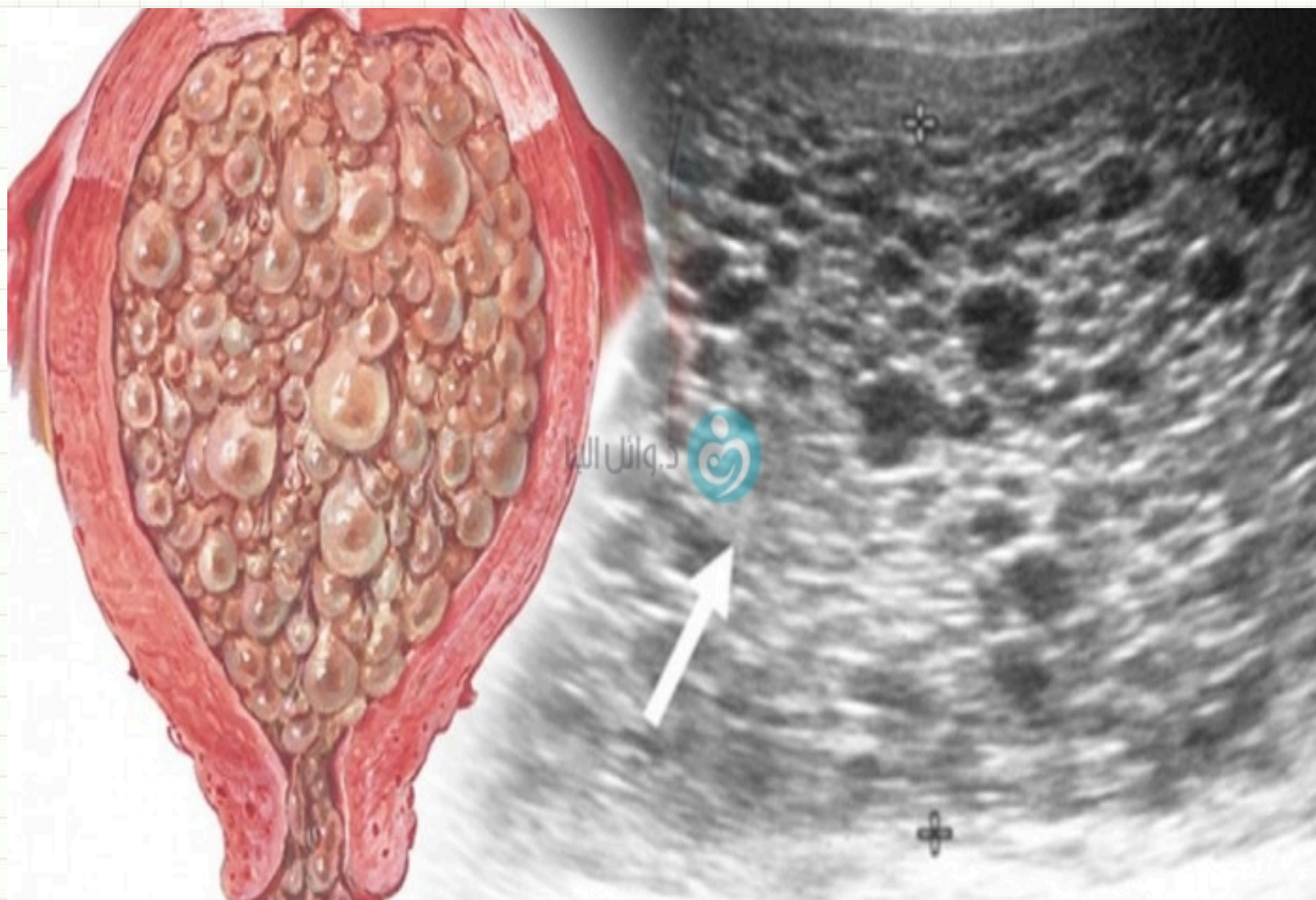
Macroscopic Often recognizable, with characteristic grape-like structures

Microscopic Diffusely hydropic villi Atypical and hyperplastic Trophoblast

Usually diagnosable from uterine products

Karyotype Usually diploid (paternally derived)

Complete moles are usually diploid, the most common type being 46XX; in most cases, a haploid sperm divides within an ovum without a nucleus.





Partial mole

Macroscopic

Can resemble hydropic abortion; may have recognizable fetal tissues

Microscopic

Focal hydropic swelling of villi

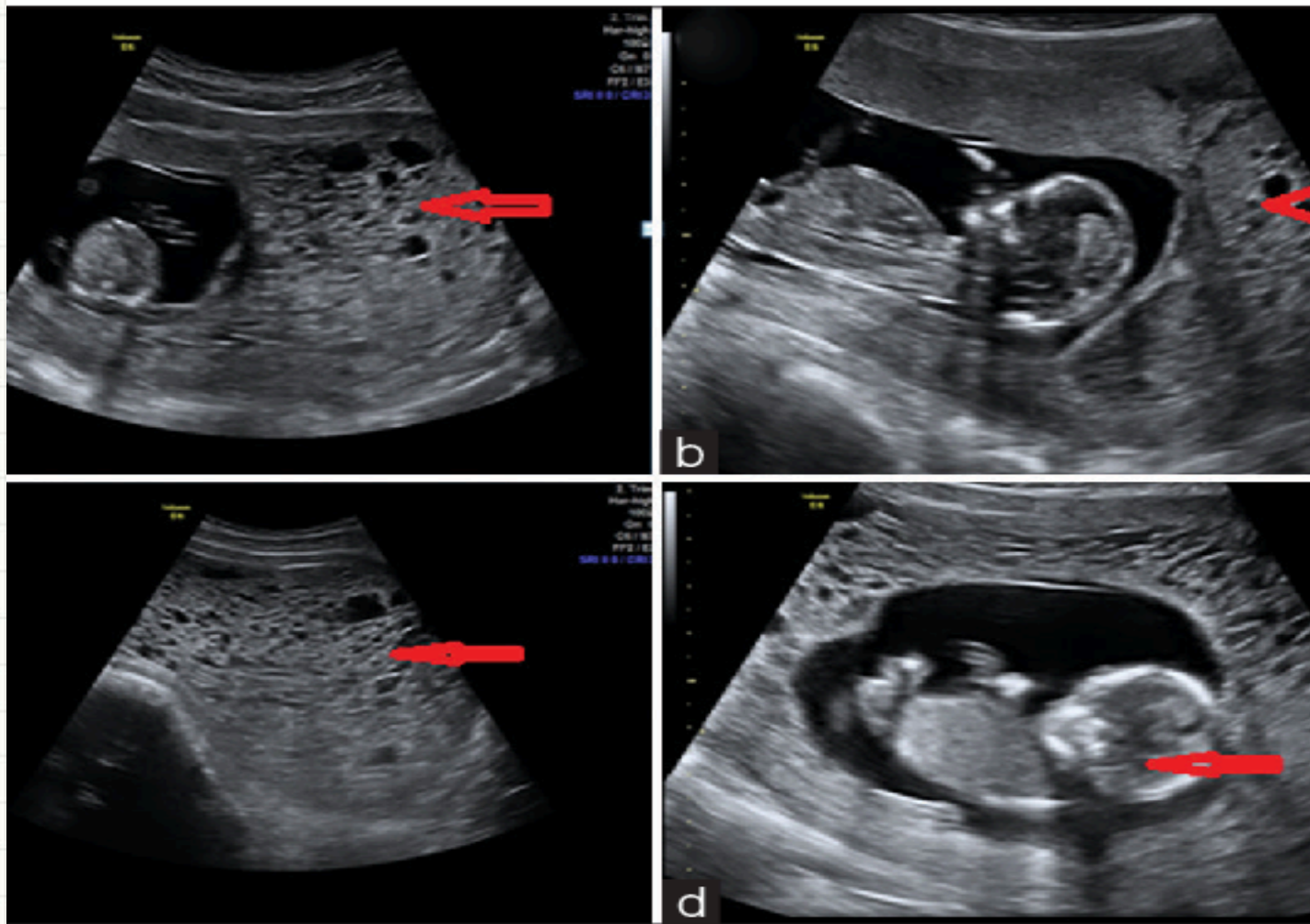
Focal trophoblastic Hyperplasia Often misdiagnosed as hydropic abortion or complete mole

Karyotype Usually triploid (maternal contribution)

partial moles usually have recognizable embryonic and fetal tissues, with focal hydropic swelling of the chorionic villi and focal trophoblastic hyperplasia.

Partial moles are generally triploid, for example 69XXY; they result most often from dispermic fertilization of normal ova.

When a fetus is present it often has the features of triploidy, including growth retardation and multiple congenital malformations



1: (a-c) Mole hydatidiform. (d) Live fetus



CLINICAL FEATURES

The majority of patients with CHM are diagnosed prior to 16 weeks of gestation, or often earlier nowadays. Abnormal vaginal bleeding during pregnancy is the usual means of presentation and this may be associated with anaemia.



CLINICAL FEATURES

Complete mole

Features May be severe and/or accompanied by paraneoplastic sequelae

Diagnosis Usually suspected on clinical or ultrasound scan findings

Persistent trophoblastic disease

In up to 20% of cases



CLINICAL FEATURES

partial mole

Features Often mild, resembling spontaneous miscarriage

Diagnosis Often unsuspected and retrospectively diagnosed after uterine evacuation

Persistent trophoblastic disease

In <0.5% of cases



DIAGNOSTIC INVESTIGATIONS

- The increasing use of ultrasound in early pregnancy has probably led to earlier diagnosis of moles.
- . In complete mole, a 'classic' pattern is seen, consisting of multiple small sonolucencies representing the numerous hydropic villi.
- In half of cases the diagnosis of partial mole is not made by ultrasound.
- In locally invasive moles, increased areas of echogenicity are seen within the myometrium.
- Transvaginal ultrasound has allowed very early diagnosis in some cases.

DIAGNOSTIC INVESTIGATIONS

Trophoblastic disease is virtually unique in that it produces a specific marker (hCG) which can be measured in urine and/or blood and correlates precisely with the amount of disease present

- The alpha subunit is similar to those of other pituitary glycoprotein hormones, but the beta subunit is specific to hCG alone.
- Higher than normal levels of hCG (particularly when $>200\,000$ IU/L) are suggestive of molar pregnancy, although levels with partial mole are only infrequently above the range for normal pregnancy



EVACUATION OF MOLAR PREGNANCY.

- Suction curettage is the method of choice for evacuation of complete molar pregnancies a suction catheter of up to 12 mm is usually sufficient.

Sharp curettage is now not generally recommended because of the possibility of uterine perforation and of increasing the risk of Asherman's syndrome (uterine synechiae).

Medical termination

- of complete mole should be avoided where possible.
- There is a theoretical concern about the routine use of potent oxytocic agents because of the possibility of forcing trophoblastic tissue into the venous spaces of the placental bed and disseminating disease to the lungs.

It is recommended that,

- when necessary, oxytocic therapy is only commenced once evacuation is complete.



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- Patients with the following should be registered:
 - complete hydatidiform mole,
 - partial hydatidiform mole,
 - twin pregnancy with complete or partial hydatidiform mole,
 - limited macroscopic or microscopic molar change judged to require follow up,
 - choriocarcinoma, placental-site trophoblastic tumour, and atypical placental-site nodules.



Persistent trophoblastic disease (gestational trophoblastic neoplasia)

- occur in up to 10 per cent in the UK
- trophoblastic disease persists, as evidenced by continuing clinical symptoms (particularly vaginal bleeding) and/or
- elevation of hCG levels. Excessive uterine size, markedly elevated hCG levels and prominent theca lutein cysts may predict persistent trophoblastic disease



Persistent trophoblastic disease (gestational trophoblastic neoplasia)

Defining persistent trophoblastic disease and the need for chemotherapy

Definitions

- High, static or rising hCG levels after one or two uterine evacuations
- Persistent uterine haemorrhage with raised hCG levels
- Pulmonary metastases with static or rising hCG levels
- Metastases in liver, brain or gastrointestinal tract
- Histological diagnosis of choriocarcinoma or placental site trophoblastic tumour

Risk factors in patients requiring chemotherapy

- >40 years old
- Increasing time since evacuation/diagnosis
- Increasing level of human chorionic gonadotrophin
- Postpartum (or unknown origin)
- Increasing number of metastases
- Increasing size of metastases
- Metastases other than in the lungs
- Previous chemotherapy

CHOICE OF CHEMOTHERAPY

- Once the decision has been made to initiate chemotherapy, the most appropriate regimen is chosen by assessing the patient's prognostic risk. This involves using a number of factors from the history, examination and investigations to allow the patient to be assigned to a risk group, which in turn facilitates the selection of the least toxic, most effective treatment for that individual

CHOICE OF CHEMOTHERAPY

- **Low risk**
- Methotrexate i.m. (low dose) or dactinomycin i.v.
- **High risk**
- Etoposide/methotrexate/dactinomycin cyclophosphamide/ vincristine (EMA-CO) (i.v. cyclical, alternating) or methotrexate–etoposide/ dactinomycin (M-EA) (i.v. cyclical, alternating)
- As GTD is highly chemosensitive, the need for surgical intervention once the diagnosis has been established is small. At present, there are essentially two further indications for hysterectomy, 1. to control severe uterine haemorrhage and 2. to eliminate disease that is confined to the uterus and resistant to chemotherapy

Indications for hysterectomy in the management of GTD

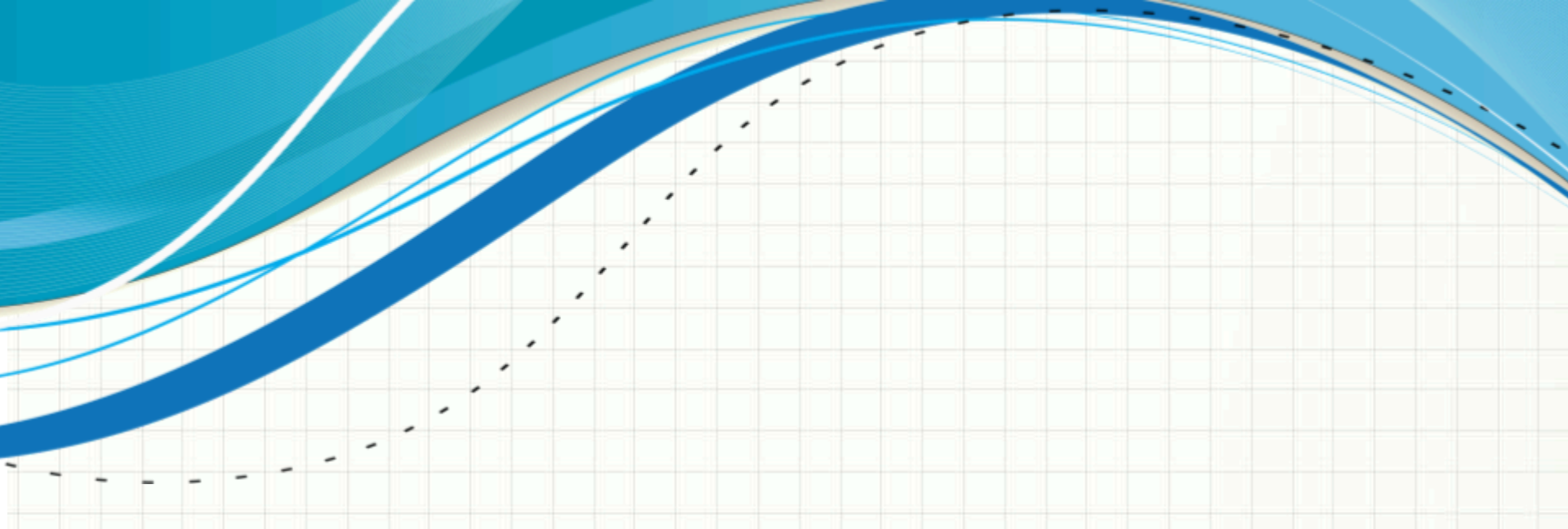
- ● Choice (older patient, localized disease, family complete).
- ● Excessive uterine bleeding (before or during treatment).
- ● Chemoresistant (localized) uterine tumour.
- ● Placental site trophoblastic tumour.

ROUTINE FOLLOW UP

- The risk of molar problems is greatest in the first 12 months following diagnosis, although it is sometimes difficult to decide whether an increase in hCG levels during monitoring is recurrence of the original or occurrence of a second mole
- ask patients to avoid further pregnancy for six months, or longer, following the molar pregnancy to enable efficient hCG follow up.

KEY POINT

- ● Registration of any molar pregnancy is essential.
- ● Ultrasound has limited value in detecting partial molar pregnancies.
- ● In twin pregnancies with a viable fetus and a molar pregnancy, the pregnancy can be allowed to proceed, after appropriate counselling.
- ● Surgical evacuation of molar pregnancies is advisable.
- ● Routine repeat evacuation after the diagnosis of a molar pregnancy is not warranted.
- ● The combined oral contraceptive pill and hormone replacement therapy are safe to use after hCG levels have reverted to normal.
- ● Women should be advised not to conceive until the hCG level has been normal for six months or follow up has been completed (whichever is sooner).



Questions?